# Erin Zaroukian

*Curriculum Vitae*

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## Positions & Education

U.S. Army Research Laboratory, Aberdeen Proving Ground, MD

Sept 2018–present **Cognitive Scientist, Computational and Information Sciences Directorate**.

Dec 2017–Sept 2018 **Post-doctoral fellow, Computational and Information Sciences Directorate**.

Oct 2014–Dec 2017 **Post-doctoral fellow, Human Research and Engineering Directorate**.

École Normale Supérieure, Paris, France

Oct 2013–Sept 2014 **Post-doctoral researcher, Laboratoire de Sciences Cognitives et Psycholinguistique**.

Johns Hopkins University, Baltimore, MD

Jan 2013–Aug 2013 **Lecturer, Center for Language Education**.

2010–2013 **Ph.D., Cognitive Science**.
Thesis: *Quantification and (un)certainty* [PDF]
Advisor: Kyle Rawlins

2007–2009 **M.A., Cognitive Science**.
Thesis: *Form, Meaning, and Comprehension/Production Asymmetries in Agreement* [PDF]
Advisor: Geraldine Legendre

2003–2007 **B.M., Trumpet**.

## Publications

2019 **Zaroukian, E.**, S. Rodriguez, S. L. Barton, J. A. Schaffer, B. Perelman, N. R. Waytowich, B. Hoffman, D. E. Asher. Algorithmically identifying strategies in multi-agent game-theoretic environments. In *Proceedings of the Society of Photo-Optical Instrumentation Engineers (SPIE): Defense + Commercial Sensing.* [LINK]

2019 Asher, D. E., S. L. Barton, **E. Zaroukian**, and N. R. Waytowich. Effect of cooperative team size on adversarial coordination with adaptive multi-agent systems. In *Proceedings of the Society of Photo-Optical Instrumentation Engineers (SPIE): Defense + Commercial Sensing.*

2019 Caylor, J., D. E. Asher, S. L. Barton, and **E. Zaroukian**. Classification of military occupation specialty codes for agent learning in human-agent teams. In *Proceedings of the Society of Photo-Optical Instrumentation Engineers (SPIE): Defense + Commercial Sensing*.

2019 **Zaroukian, E.**, J. Caylor, M. Vanni, and S. Kase. Evaluating Improvement in Situation Awareness and Decision-Making Through Automation. In *Proceedings of the IEEE Conference on Cognitive and Computational Aspects of Situation Management (CogSIMA)*.

2019 Barton, S., **E. Zaroukian**, D. Asher, and N. Waytowich. Evaluating the Coordination of Agents in Multi-Agent Reinforcement Learning. In *Proceedings of the International Conference on Intelligent Human Systems Integration (IHSI)*, Ed. W. Karwowski, T. Ahram, 765-770. [LINK]

2019 Barton, S., N. Waytowich, **E. Zaroukian**, and D. Asher. Measuring collaborative emergent behavior in multi-agent reinforcement learning. In *Proceedings of the International Conference on Human Systems Engineering and Design (IHSED): Future Trends and Applications*, Ed. W. Karwowski, T. Ahram, 422-427. [LINK] [LINK]

2018 **Zaroukian, E.** Information extraction for optimized human understanding and decision making. In *Proceedings of the 23rd International Command and Control Research and Technology Symposium (ICCRTS)*.* [PDF]

2018 Barton, S., N. Waytowich, **E. Zaroukian**, and D. Asher. Measuring collaborative emergent behavior in multi-agent reinforcement learning. In *Proceedings of the 23rd International Command and Control Research and Technology Symposium (ICCRTS)*.* [PDF]

2018 **Zaroukian, E.**, and J. Bakdash. Human understanding of information represented in natural versus artificial language. In *IEEE International Inter-Disciplinary Conference on Cognitive Methods in Situation Awareness and Decision Support (CogSIMA)*. IEEE.* [PDF] [LINK]

2018 Bakdash, J., C. Sample, M. Rankin, M. Kantarcioglu, J. Holmes, S. Kase, **E. Zaroukian**, and B. Szymanski. The Future of Deception: Machine-Generated and Manipulated Images, Video, and Audio?. In *2018 International Workshop on Social Sensing (SocialSens)*, 2-2. [LINK]

2017 Preece, A., W. Webberley, D. Braines, **E. Zaroukian**, and J. Bakdash. SHERLOCK: Experimental Evaluation of a Conversational Agent for Mobile Information Tasks. In *IEEE Transactions on Human-Machine Systems*, 1-12.* [LINK]

2017 Preece, A., I. Taylor, A. Dawson, D. Braines, N. O'Leary, A. Thomas, R. Tomsett, T. La Porta, J. Z. Bakdash, and **E. Zaroukian**. Coalitions of things: supporting ISR tasks via internet of things approaches. In *Proceedings of SPIE 10190, Ground/Air Multisensor Interoperability, Integration, and Networking for Persistent ISR VIII*, Ed. T. Pham and M. A. Kolodny, 101900A–101900A-9. [LINK]

2017 **Zaroukian, E.**, J. Bakdash, A. Preece, and W. Webberley. Automation Bias with a Conversational Interface. In *IEEE International Inter-Disciplinary Conference on Cognitive Methods in Situation Awareness and Decision Support (CogSIMA)*. IEEE.* [PDF]

2016 Preece, A., W. Webberley, D. Braines, **E. Zaroukian**, and J. Bakdash. Human Computer Collaboration at the Edge: Enhancing Collective Situation Understanding with Controlled Natural Language. In *Proceedings of the 21st International Command and Control Research and Technology Symposium (ICCRTS)*.* [PDF]

– Awarded best paper in the track *Socio-technical Networks in Connected Battlespace*

2016 **Zaroukian, E.** Human Understanding of Controlled Natural Language in Simulated Tactical Environments. In *IEEE International Inter-Disciplinary Conference on Cognitive Methods in Situation Awareness and Decision Support (CogSIMA)*, 126-130. IEEE.* [PDF]

2016 Legendre, G., M. Putnam, H. de Swart, and **E. Zaroukian** (Eds). Optimality-theoretic syntax, semantics & pragmatics: From uni- to bidirectional optimization. Vol. 61. Oxford University Press. [LINK]

2015 Mott, D. and **E. Zaroukian** Studies in the Human Use of Controlled English. In *Annual Fall Meeting of the International Technology Alliance*. [PDF]

2015 Preece, A., W. Webberley, D. Braines, N. Hu, T. La Porta, **E. Zaroukian**, and J. Z. Bakdash. SHERLOCK: Simple Human Experiments Regarding Locally Observed Collective Knowledge. In *Annual Fall Meeting of the International Technology Alliance*. [PDF]

2014 **Zaroukian, E.** (*Just*) *about*: an analysis. In *Proceedings of the 43rd annual meeting of the North East Linguistic Society Volume Two*, Ed. H-L Huang, E. Poole, A. Rysling, 273-285. [PDF]

2014 Legendre, G., J. Culbertson, **E. Zaroukian**, L. Hsin, I. Barrière, and T. Nazzi. Is children's comprehension of subject-verb agreement universally late? Comparative evidence from French, English, and Spanish. In *Lingua (Special Issue), How children learn to detect and interpret agreement morphology: A cross-linguistic perspective*, Ed. A. Pérez-Leroux, 21-39.* [PDF]

2014 **Zaroukian, E.** and C. Beller. Licensing by modification in two classes of verbs. In *Proceedings of the 48th Annual Meeting of the Chicago Linguistic Society*, 619-633. [PDF]

2013 **Zaroukian, E.** and C. Beller. Not all *have*-clauses are alike. In *Proceedings of Sinn und Bedeutung 17 (online)*, ed. E. Chemla, V. Homer, and G. Winterstein, 635-652. Paris: ENS Paris. [PDF]

2012 **Zaroukian, E.** and C. Beller. Evaluating singular indefinites. In *Proceedings of the 40th Western Conference on Linguistics*, Ed. C. Galeano, Görgülü, E., Presnyakova, I., 127-138. California State University at Fresno. [PDF]

2012 **Zaroukian, E.** Gradable predicates and the distribution of approximators. In *Proceedings of the 40th Western Conference on Linguistics*, Ed. C. Galeano, Görgülü, E., Presnyakova, I., 115-126. California State University at Fresno. [PDF]

2012 Beller, C. and **E. Zaroukian**. Stage-level evaluativity is desiderativity. In *Proceedings of Sinn und Bedeutung 16*, ed. A. Aguilar-Guevara, A. Chernilovskaya, and R. Nouwen, 71-84. Cambridge, MA: MIT Working Papers in Linguistics. [PDF]

2012 **Zaroukian, E.** Approximative Inversion Revisited. In *Proceedings of the 19th annual workshop on Formal Approaches to Slavic Linguistics*, ed. J. F. Bailyn, E. Dunbar, Y. Kronrod, and C. LaTerza, 146-160. Ann Arbor, MI: Michigan Slavic Publications.* [PDF]

2012 **Zaroukian, E.** Homophony and Russian copular constructions. In *Proceedings of the 8.5th European Conference on Formal Description of Slavic Languages*, ed. M. Ziková and M. Dočekal, 255-268. Frankfurt: Peter Lang.* [PDF]

2011 **Zaroukian, E.** Divergent approximators. *In Proceedings of Sinn und Bedeutung 15*, ed. I. Reich, E. Horch, and D. Pauly, 677-690. Saarbrücken, Germany: Universaar – Saarland University Press. [PDF]

2007 Legendre, G., T. Nazzi, I. Barriere, J. Culbertson, M. Lopez-Gonzalez, L. Goyet and **E. Zaroukian**. Acquiring subject-verb agreement in French: Evidence of early syntactic representations from comprehension. In *Proceedings of the 31st Annual Boston University Conference on Language Development*, ed. H. Caunt-Nulton, S. Kulatilake, and I. Woo, 370-381. Somerville, MA: Cascadilla Press. [PDF]

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## Manuscripts

2014 Bakdash, J., S. Hutchinson, **E. Zaroukian**, L. Marusich, S. Thirumuruganathan, C. Sample, B. Hoffman, G. Das. Malware in the Future? Forecasting Analyst Detection of Cyber Events. [LINK]

2014 **Zaroukian, E.** Uncertainty, prosody, and their interaction: a case of modal concord?. [PDF]

## Conference Presentations

2019 **Zaroukian, E.**, S. S. Rodriguez, S. L. Barton, J. A. Schaffer, B. Perelman, N. R. Waytowich, B. Hoffman, D. E. Asher. Algorithmically identifying strategies in multi-agent game-theoretic environments. Paper presented at SPIE: Defense + Commercial Sensing. [SLIDES]

2019 Caylor, J., D. Asher, S. Barton, and **E. Zaroukian**. Classification of military occupation specialty codes for agent learning in human-agent teams. Paper presented at SPIE: Defense + Commercial Sensing.

2019 Asher, D., S. Barton, **E. Zaroukian**, and N. Waytowich. Effect of cooperative team size on adversarial coordination with adaptive multi-agent systems. Paper presented at SPIE: Defense + Commercial Sensing.

2019 **Zaroukian, E.**, J. Caylor, M. Vanni, and S. Kase. Evaluating Improvement in Situation Awareness and Decision-Making Through Automation. Poster presented at CogSIMA. [POSTER]

2018 Barton, S., **E. Zaroukian**, D. Asher, and N. Waytowich. Evaluating the Coordination of Agents in Multi-Agent Reinforcement Learning. Paper presented at IHSI.

2018 Barton, S., N. Waytowich, **E. Zaroukian**, and D. Asher. Measuring collaborative emergent behavior in multi-agent reinforcement learning. Paper presented at IHSED.

2018 **Zaroukian, E.** Information extraction for optimized human understanding and decision making. Paper presented at ICCRTS.

2018 Barton, S., N. Waytowich, **E. Zaroukian**, and D. Asher. Measuring collaborative emergent behavior in multi-agent reinforcement learning. Paper presented at ICCRTS.

2018 **Zaroukian, E.** Deductive reasoning with a simulated information extraction pipeline. Poster presented at AHFE.

2018 **Zaroukian, E.**, and J. Bakdash. Human understanding of information represented in natural versus artificial language. Poster presented at CogSIMA. [POSTER]

2018 Bakdash, J., C. Sample, M. Rankin, M. Kantarcioglu, J. Holmes, S. Kase, **E. Zaroukian**, and B. Szymanski. The Future of Deception: Machine-Generated and Manipulated Images, Video, and Audio?. Paper presented at SocialSens.

2017 Preece, A., I. Taylor, D. Braines, A. Thomas, N. O'Leary, T. La Porta, J. Bakdash, and **E. Zaroukian**. Coalitions of Things: Supporting ISR tasks via Internet of Things approaches. Paper presented at SPIE Security + Defense.

2017 **Zaroukian, E.**, J. Bakdash, A. Preece, and W. Webberley. Automation Bias with a Conversational Interface. Poster presented at IEEE International Multi-Disciplinary Conference on Cognitive Methods in Situation Awareness and Decision Support (CogSIMA). [POSTER]

2016 Preece, A., W. Webberley, D. Braines, **E. Zaroukian**, and J. Bakdash. Collective Situation Understanding with Controlled Natural Language. Paper presented at the International Command and Control Research and Technology Symposium (ICCRTS).

2016 **Zaroukian. E.** Human Understanding of Controlled Natural Language in Simulated Tactical Environments. Poster presented at IEEE International Multi-Disciplinary Conference on Cognitive Methods in Situation Awareness and Decision Support (CogSIMA). [POSTER]

2015 Tieu, L. and **E. Zaroukian** 'Sorta' hedging objects, verbs, and whole sentences. Poster presented at Experimental Pragmatics (XPRAG). [POSTER]

2015 **Zaroukian, E.** and L. Tieu. Hedging arguments. Poster presented at the 89th annual meeting of the Linguistic Society of America (LSA). [EXTENDED ABSTRACT]

2014 **Zaroukian, E.** Unified by degrees. Poster presented at the 88th annual meeting of the Linguistic Society of America (LSA). [POSTER]

2014 **Zaroukian, E.** and B. Beller. The strong and the weak: evaluation in modal desideratives. Paper presented at the 88th annual meeting of the Linguistic Society of America (LSA). [HANDOUT]

2013 **Zaroukian, E.** *Approximately* and *about*: epistemic possibility in approximation. Poster presented at the third Mid-Atlantic Colloquium of Studies in Meaning (MACSIM). [POSTER]

2013 Beller, C., K. Johannes, M. Oliver, **E. Zaroukian**, and K. Rawlins. Scope and agent-orientation in adverbs. Poster presented at the 87th annual meeting of the Linguistic Society of America (LSA). [POSTER]

2012 **Zaroukian, E.** (*Just*) *about*: an analysis. Poster presented at the 43rd meeting of the North East Linguistic Society (NELS). [POSTER] [HANDOUT]

2012 **Zaroukian, E.** and C. Beller. Not all *have*-clauses are alike. Poster presented at Sinn und Bedeutung 17. [POSTER] [HANDOUT]

2012 **Zaroukian, E.** and C. Beller. Licensing by modification in two classes of verbs. Paper presented at the 48th Annual Meeting of the Chicago Linguistic Society (CLS). [HANDOUT]

2012 **Zaroukian, E.** Evaluating quantities. Poster presented at the second Mid-Atlantic Colloquium of Studies in Meaning (MACSIM). [POSTER]

2012 Beller, C. and **E. Zaroukian**. Evaluation and consumption. Paper presented at the 86th annual meeting of the Linguistic Society of America (LSA). [HANDOUT] [EXTENDED ABSTRACT]

2012 **Zaroukian, E.** Approximation and the coercion of gradable predicates. Poster presented at the 86th annual meeting of the Linguistic Society of America (LSA). [POSTER] [EXTENDED ABSTRACT]

2011 **Zaroukian, E.** and C. Beller. Evaluating singular indefinites. Paper presented at the Western Conference on Linguistics (WECOL). [HANDOUT]

2011 **Zaroukian, E.** Gradable predicates and the distribution of approximators. Poster presented at the Western Conference on Linguistics (WECOL). [POSTER]

2011 Beller, C. and **E. Zaroukian**. Stage-level evaluativity is desiderativity. Paper presented at Sinn und Bedeutung 16. [HANDOUT]

2011 **Zaroukian, E.** Uncertainty and rising intonation. Paper presented at the 85th annual meeting of the Linguistic Society of America (LSA). [HANDOUT] [EXTENDED ABSTRACT]

2010 **Zaroukian, E.** Homophony and Russian copular constructions. Paper presented at the 8.5th European Conference on Formal Description of Slavic Languages (FDSL). [HANDOUT]

2010 **Zaroukian, E.** Divergent approximators. Paper presented at Sinn und Bedeutung 15. [HANDOUT]

2010 **Zaroukian, E.** What are Uncertain Numerals?. Poster presented at the 2010 IGERT Project Meeting. [POSTER]

2010 **Zaroukian, E.** Approximative Inversion Revisited. Paper presented at the 19th annual workshop on Formal Approaches to Slavic Linguistics (FASL). [HANDOUT]

2010 **Zaroukian, E.** Uncertain numerals. Poster presented at the first Mid-Atlantic Colloquium of Studies in Meaning (MACSIM). [POSTER]

2010 **Zaroukian, E.** Phrasal homophony: an interaction between syntax and phonology. Paper presented at the meeting of the Georgetown Linguistics Society (GLS). [HANDOUT]

2010 **Zaroukian, E.** Expressing numerical uncertainty. Paper presented at the 84th annual meeting of the Linguistic Society of America (LSA). [HANDOUT] [EXTENDED ABSTRACT]

## Invited talks

2012 **Zaroukian, E.** Concord in contingent commitments?. Talk given at the CUNY Syntax Supper. [HANDOUT]

## Teaching

### Instructor, Johns Hopkins University

Summer 2013 **Semantics I**, *Cognitive Science Department*.
Co-instructor: Mike Oliver

Summer 2013 **Listening & Speaking**, *Center for Language Education*.

Spring 2013, Summer 2013 **Accent Reduction**, *Center for Language Education*.

Intersession 2011, 2012, 2013 **Inventing Language**, *Cognitive Science Department*.

### Teaching Assistant, École Normale Supérieure

Spring 2014 **Formal Semantics**, *Institut d'Étude de la Cognition*.
Instructors: Emmanuel Chemla and Vincent Homer

### Teaching Assistant, Johns Hopkins University

Fall 2012 **Semantics I**, *Cognitive Science Department*.
Instructor: Kyle Rawlins

Spring 2011 **Cognitive Neuroscience**, *Cognitive Science Department/Neuroscience*.
Instructor: Brenda Rapp

Spring 2011 **World of Language**, *Cognitive Science Department*.
Instructor: Geraldine Legendre

Fall 2010 **Intro to Cognitive Development**, *Cognitive Science Department*.
Instructor: Barbara Landau

Spring 2010 **Language and Advertising**, *Cognitive Science Department*.
Instructor: Kyle Rawlins

Fall 2009 **Syntax I**, *Cognitive Science Department*.
Instructor: Geraldine Legendre

Spring 2009 **Language & Mind**, *Cognitive Science Department*.
Instructor: Joan Chen-Main

Fall 2008 **Formal Methods in Cognitive Science: Language**, *Cognitive Science Department*.
Instructor: Kyle Rawlins

Spring 2008 **Phonology I**, *Cognitive Science Department*.
Instructor: Colin Wilson

## Grants, Fellowships,and Awards

2012 Gaston Sweitzer Fellowship recipient, Johns Hopkins University

2007–2012 IGERT trainee, grant awarded to JHU Department of Cognitive Science

2007 Linguistic Society of America Summer Institute fellowship

2007 Azalia H. Thomas prize for distinguished work in music theory, Johns Hopkins University

2007 NSF Research Experiences for Undergraduates program awardee

2004 North American First Year Student Scholarship, Freie Universität

2003–2007 Peabody Merit Scholarship, Johns Hopkins University

## Service

### Reviewer

2015 **Linguistic Society of America 90th annual meeting**.

2014 **Linguistic Society of America 89th annual meeting**.

2014 **Formal Description of Slavic Languages 10**.
University of Leipzig

2012 **Johns Hopkins University Optimality Theory Workshop**.
Cognitive Science Department, Johns Hopkins University

### Organizer

2011–2012 **Johns Hopkins University Optimality Theory Workshop**.
Cognitive Science Department, Johns Hopkins University

2013 **Mid-Atlantic Colloquium of Studies in Meaning**.
Cognitive Science Department, Johns Hopkins University

2012 **Johns Hopkins University IGERT Conference**.
Cognitive Science Department, Johns Hopkins University

### Editor

2012–present **Johns Hopkins University Optimality Theory Workshop**.
Cognitive Science Department, Johns Hopkins University

### Committee member

2010–2013 **Colloquium committee**.
Cognitive Science Department, Johns Hopkins University

2010–2012 **Travel Grant committee**.
Graduate Representative Organization, Johns Hopkins University

### Representative

2008–2013 **Department representative**.
Graduate Representative Organization, Johns Hopkins University

### Mentor

2008-2011 **GradNet mentoring program**.
Johns Hopkins University

## Technical skills

Experimental design and statistical analysis in SPSS, MATLAB, PsychoPy, and R

Programming in C++, Java, Python, JavaScript, HTML, MATLAB

Text formatting in LaTeX

## Languages

| | |
|---|---|
| English | Native |
| German | Proficient |
| Russian | Beginning |
| French | Beginning |

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